Journal of Quantitative Spectroscopy & Radiative Transfer 353 (2026) 109854

Contents lists available at ScienceDirect

Journal of Quantitative Spectroscopy and Radiative Transfer

journal homepage: www.elsevier.com/locate/jqsrt

Beyond-Voigt collisional parameters of subTHz H_2O lines for atmospheric applications☆

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ARTICLE INFO

Keywords:
Microwave spectroscopy
Water
Line shape analysis
Collisional parameters
Speed dependence effect
Atmospheric applications

ABSTRACT

The profiles of nine key subTHz H_2O lines were recorded at various pressures and room temperature using two fundamentally different spectroscopic techniques. The beyond-Voigt line shape taking into account the speed-dependence of the collision relaxation rate was employed to analyze the high quality experimental recordings. Collisional broadening, shifting and their quadratic speed-dependence were measured for major atmospheric perturbers H_2O, N_2, O_2, and Ar. The found air-related line shape parameters can be used for updating atmospheric databases and propagation models.

Introduction

Water vapor plays a critical role in atmospheric processes, being the main absorber and primary greenhouse gas and influencing numerous radiative and chemical interactions. Water vapor absorbs electromagnetic waves ranging from radio to ultraviolet, with the frequency region below 1 THz being particularly valuable for water vapor sounding. The shape of the key resonant lines located in this range and their sensitivity to environmental conditions provide essential data for atmospheric remote sensing and climate modeling [1]. Knowing precise parameters of the shape of subTHz H_2O lines — particularly, collisional broadening, shifting, and their speed-dependence — forms a crucial foundation for atmospheric remote sensing and radiative transfer modeling. With the improvement of measurement techniques and computational models, these spectroscopic insights will further enhance the potential of atmospheric water vapor monitoring with increasing precision, ultimately contributing to more accurate climate projections and weather forecasting.

Despite the long history of studying water molecule lines in the microwave region (see, for example, [2] and references therein), speed-dependent line parameters are measured only for the lines at 22 [3] and 183 GHz [4]. Quite a few datasets are available on the collisional broadening and shifting coefficients of the water molecule lines in the range of 300–1000 GHz within the speed-independent approach [5–10]. HITRAN [11] uses these sources in combination with the recalculations and semi-empirical results from the experimental and theoretical studies of rovibrational lines [12–14]. The database contains information on the self- and air-broadening and air-shifting, while self-shifting coefficients are not provided. In the study of the lines above 1 THz [15], the list of subTHz lines' self-broadening and self-shifting coefficients obtained by means of semi-classical calculations is given as a supplementary data. All these data suggest using a speed-independent profile, limiting the accuracy of line shape modeling. In addition, speed-dependence broadening coefficients for a number of lines in the range 1.0–1.2 THz [15–17] are known.

A number of radiative transfer models adapts sets of the molecular lines' parameters for radiometric studies of the atmosphere. In the sub-THz range, the PyRTlib library [18] allows modeling radiometric parameters (e.g., brightness temperature) based on the meteorological data. It includes various atmospheric air absorption models (for different line shapes) with the parameters updated as new data become available. The most recent version accounts for the contributions of the spectral lines up to 1100 GHz. The library provides SD-parameters only for the H_2O line near 183 GHz [4] due to the lack of measurements for the other lines.

The current study aims to improve the knowledge of the collisional parameters of the water molecule subTHz lines beyond the Voigt profile considering the speed dependence of the collisional cross section essential for atmospheric applications.

The paper is organized as follows: in Section 1 the measurement techniques are described and the approaches to processing the obtained

☆ This article is part of a Special issue entitled: 'Memorial issue in honor of Mikhail Tretyakov' published in Journal of Quantitative Spectroscopy and Radiative Transfer.

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https://doi.org/10.1016/j.jqsrt.2026.109854
Received 22 December 2025; Received in revised form 26 January 2026; Accepted 26 January 2026
Available online 28 January 2026
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water line recordings are discussed. Section 2 focuses on the line shape parameter retrieval from experimental recordings and on the analysis of the pressure-dependence of collisional parameters. The accuracy of the atmospheric absorption modeling with retrieved parameters against the widely used models referred to broadband measurement data under atmospheric conditions is discussed in Section 3. The summary of the study is given in Section 4.

1. Measurements

The profiles of the water molecule lines in the 325–860 GHz region were recorded at room temperature and various pressures of pure water vapor and its mixtures with N_2, O_2 and Ar using a video spectrometer (VS) and a spectrometer with radioacoustic detector of gas absorption (RAD spectrometer). The experimental line shape parameters were verified using the humid air absorption spectrum recorded by means of a resonator spectrometer.

A series of backward wave oscillators (BWOs) covering the studied frequency range were employed as a radiation source in all spectrometers. Radiation frequency was stabilized by means of the phase-locked loop over a harmonic of the microwave frequency synthesizer which allowed precise frequency control and spectrum recording by step-by-step scanning of the synthesizer frequency.

The video spectrometer [19,20] was used to obtain line profile recordings at pressures of the studied gas mixture (in our case, pure water vapor or one mixed with oxygen, nitrogen, or argon) within 50–600 mTorr. Cylindrical gas cells with a length of 73 cm or 134 cm and 10 cm in diameter, depending on the studied line intensity, were used. Line profiles were recorded using the modulation of the radiation power (also called amplitude modulation and denoted as AM) with 190 Hz modulation frequency and further the demodulation of the output signal by a phase-lock detector. The radiation that had passed through the cell was detected with a liquid helium cooled InSb hot electron bolometer. Typically, each spectrum recording contained 501 frequency points. The temperature was monitored by several Pt100 sensors mounted on the cell walls: the temperature was stable within ±0.5 K during a pressure ramp experimental cycle; day to day variations were within 298–300 K.

The RAD spectrometer [21–24] was used for recording the lines in the 50–2500 mTorr pressure range of the studied gas mixture at room temperature (the cell temperature was stabilized by the active thermostat within ~0.1 K). Each recording contained 151 frequency points. The gas cell lengths were chosen depending on the studied line intensity to minimize saturation due to the large optical depth: from 10 cm for the weakest line at 859 GHz and down to 1 cm for the most intense ones. The use of the radiation frequency manipulation, FMp (a square-wave frequency modulating function with a precisely known and fixed deviation) and further demodulation of the spectrometer output signal by means of a lock-in amplifier with reference frequency equal to the manipulation frequency allowed increasing the spectra quality and significantly reducing the spectrometer baseline. To achieve a sufficient signal-to-noise ratio, several (from 4 to 24) multiple single scans (0.5 s time constant per frequency point) were recorded under permanent conditions and then averaged.

Both with VS and RAD spectrometers, data were obtained in batches as follows. At first, the pre-evacuated gas cell was filled with some amount of H_2O vapor and the line profile was recorded. Then a portion of H_2O (for self-broadening study) or buffer gas was added step by step (~40 mTorr for the video spectrometer, ~300 mTorr for the RAD spectrometer) and the line profile was recorded after achieving chemical equilibrium (in about half an hour). The latter was controlled by monitoring the time variation of the line peak absorption. For each perturbing molecule, the recording batches were generally repeated several (up to 7) times at different relative positions of the radiation source and the gas cell to estimate (and average) the influence of the standing waves on the measured line parameters. The total number of the recordings obtained within the study is specified in Table 1 together with the information on frequencies, intensities and quantum numbers of the studied lines. For the most intense lines ($J'_{K'_a,K'_c} \leftarrow J_{K_a,K_c} = 1_{10} \leftarrow 1_{01}$ at 556.9 GHz and $2_{11} \leftarrow 2_{02}$ at 752 GHz), the self-broadening and self-shifting parameters were hard to measure with the video spectrometer due to the large optical depth. Therefore, the lines of the $H_2^{18}O$ isotopologue corresponding to the transitions with the same quantum numbers having intensities of $1.008 \cdot 10^{-22}$ and $6.715 \cdot 10^{-23}$ respectively were studied assuming a minor isotopological effect on the collisional line shape parameters.

Table 1
Summary on the studied lines and total amount of the analyzed line profile recordings made with video and RAD spectrometers (frequencies and intensities are from HITRAN database [11]).

Frequency (MHz)	Intensity (cm/molec.)	Transition assignment $J'_{K'_a,K'_c} \leftarrow J_{K_a,K_c}$	Recordings VS	Recordings RAD
325152.937	$9.074 \cdot 10^{-23}$	$5_{15} \leftarrow 4_{22}$	473	182
380197.360	$8.259 \cdot 10^{-22}$	$4_{14} \leftarrow 3_{21}$	1318	300
439150.864	$7.126 \cdot 10^{-23}$	$6_{43} \leftarrow 5_{50}$	394	82
448001.128	$8.630 \cdot 10^{-22}$	$4_{23} \leftarrow 3_{30}$	283	82
474689.166	$1.090 \cdot 10^{-22}$	$5_{33} \leftarrow 4_{40}$	292	137
556936.069	$5.236 \cdot 10^{-20}$	$1_{10} \leftarrow 1_{01}$	176	77
620701.035	$5.670 \cdot 10^{-22}$	$5_{32} \leftarrow 4_{41}$	152	105
752033.234	$3.452 \cdot 10^{-20}$	$2_{11} \leftarrow 2_{02}$	291	142
859965.729	$1.903 \cdot 10^{-23}$	$2_{11} \leftarrow 2_{02}$ ($v_2 = 1$)	–	60

The detailed description of the resonator spectrometer used for recording the humid air absorption spectrum can be found in [25]. Briefly, gas sample absorption was determined from the change of the quality factor (found from the resonance curve width) of a Fabry–Perot resonator after its filling with the studied gas. The resonator intrinsic losses were assessed when the resonator was filled with nonabsorbing gas (e.g., argon) under pressure to keep the same eigen-mode frequencies without changes in the resonator length. Moist air absorption was recorded in the frequency range of about 350–500 GHz at room temperature (296 K) and pressure of 758.5 Torr. The water vapor partial pressure was 1.667 Torr.

2. Data treatment

The sensitivity of both video and RAD spectrometers allows observation of the beyond-Voigt line shape features. As a line shape model taking into account the speed-dependence of the collision cross section we used the Hartmann–Tran profile (hereinafter referred to as HTp) [26] that was reduced to the quadratic speed-dependent (SD) Voigt profile (qSDVp) by setting to zero the velocity-changing collisions rate ν_{vc} and the correlation parameter η. The ν_{vc} value could be derived from the fitting procedure or introduced as a fixed value based, e.g., on calculations, but for the subTHz H_2O lines neither relevant results were found, nor a reliable value from the fitting procedure could be retrieved.

The recorded spectra were treated using least-squares fitting of the model function to the experimental data employing the Levenberg–Marquardt method. For the recordings obtained with power modulation, the model function $S^{AM}(f)$ was based on the true spectral line profile $F(f)$ (such as Voigt or HTp) with a linear multiplicative term to account for the frequency variation of the radiation power and additive polynomial (up to 2nd order) term for the spectrometer baseline:

$$S^{AM}(f) = S_0 \cdot (1 + a_0(f - f_0)) \cdot F(f) + a_1 + a_2(f - f_0) + a_3(f - f_0)^2, \quad (1)$$

where f is the frequency, f_0 is the line central frequency, S_0 and $a_{0..3}$ are the variable parameters determined from the fit. In the case of frequency manipulation, the shape of the observed line is determined as the difference of two model functions $S^{AM}(f)$ shifted from the true line center f_0 by the value of deviation $\pm dev$:

$$S^{FMp}(f) = S^{AM}(f + dev) - S^{AM}(f - dev). \quad (2)$$

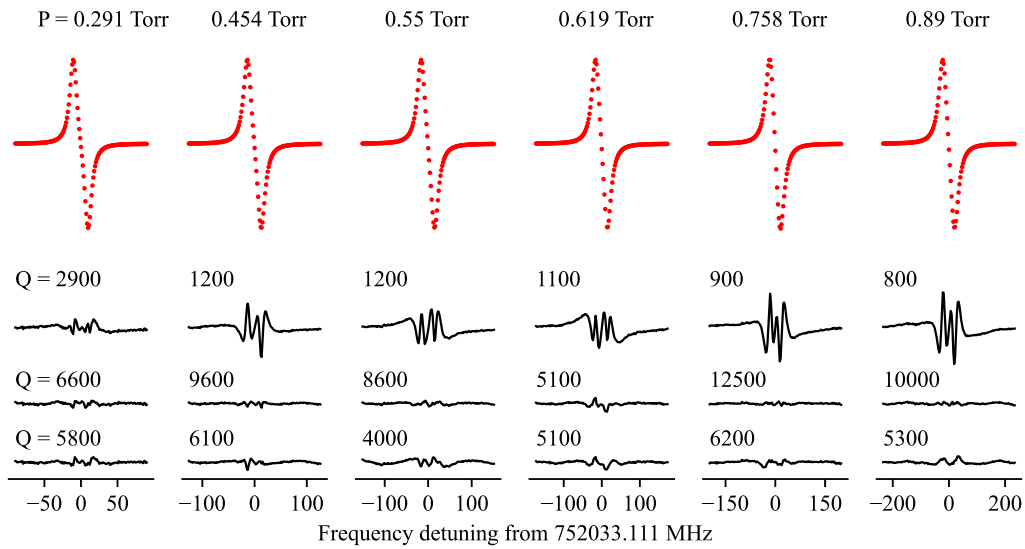

Fig. 1. A series of RAD spectrometer recordings (normalized to a single amplitude) of the self-broadened line near 752 GHz at 298.6 K (points, upper panels) and fit residuals (scaled by a factor of 100) for the speed-independent Voigt profile (upper solid traces) and qSDVp for line-by-line and multifit analysis (middle and lower solid traces, respectively). Fit quality Q (maximum signal to residual RMS ratio) is shown near each residual.

Two approaches were used for the analysis of experimental recordings: (1) line-by-line fitting with subsequent analysis of pressure dependences and (2) multispectrum fitting (or "multifit") with constraints on the linear pressure dependence of line shape parameters. More details on the data processing procedure can be found in our previous works [27,28]. The parameters of interest are central frequency f_0, speed-averaged collisional half-width Γ_0 and its quadratic SD parameter Γ_2, speed-averaged collisional shift of the central frequency Δ_0 and its quadratic SD parameter Δ_2 (the latter is rather small and could be retrieved from the fit not in all cases). The corresponding coefficients of the linear pressure dependence of the collisional line shape parameters are referred to as γ_0, γ_2, δ_0, and δ_2, respectively. As the parameters of the speed-dependent profile were retrieved for the first time, the speed-independent (SI) Voigt profile was also used for data processing to retrieve the speed-independent broadening γ_V and shifting δ_V parameters for comparison with the earlier results.

A typical example of the data analysis of the RAD spectrometer recordings of the line at 752 GHz in pure water vapor using different approaches is shown in Fig. 1. The fit residuals in the case of the Voigt profile (upper black traces) exhibit the systematic w-shaped discrepancy demonstrating that the speed-independent profile does not correctly describe the observed line shape limiting the accuracy of its modeling. The qSDV profile improves the experimental spectra description up to almost the experimental noise level. The multifit residuals, as discussed earlier [28], demonstrate a less fit quality factor due to some uncertainties of temperature and pressure readings. The latter ones are more essential for H_2O in comparison to other molecules due to its more pronounced ability to be absorbed by the surfaces of the cell and vacuum system components. It should also be noticed that no evident manifestation of the Dicke effect (velocity-changing collisions) is observed in the fit residuals at the lowest pressures in contrast to our study of the R(0) line of CO molecule (see Fig. 1 in [28]).

Figs. 2 and 3 display the pressure dependences of Γ_0, Γ_2 and Δ_0 retrieved from the video- and RAD spectrometer recordings. The points are the results of the line-by-line analysis, the straight lines correspond to the broadening and shifting coefficients γ_0, γ_2 and δ_0 retrieved from the multifit analysis of the RAD spectrometer data. The lower parts of both figures represent the difference between the experimental points and the linear dependence. The VS data are more scattered and demonstrate greater fit uncertainty (only line-by-line fitting was used due to this), but are generally in a good agreement with the RAD spectrometer data. In addition, high sensitivity of the RAD spectrometer allowed observation of the speed-dependent frequency shifting for some lines. An example of Δ_2 versus pressure dependence is given in Fig. 4 for the 752 GHz line. The values of Δ_2 are near 10% of Δ_0 (which is noticeable in the case of self-shifting) but they are quite scattered and sometimes

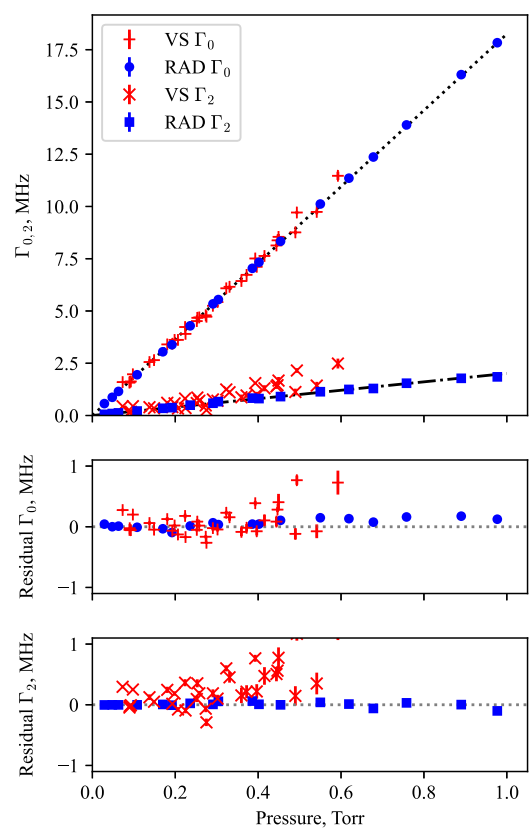

Fig. 2. Half-width Γ_0 and SD parameter Γ_2 of 752 GHz line versus water pressure (top plot). Video and RAD spectrometers datasets are shown by red and blue symbols respectively (see figure inset). Line slopes are results of multifit analysis of RAD spectrometer data. Residuals are presented in middle (Γ_0) and bottom (Γ_2) plots by the corresponding symbols.

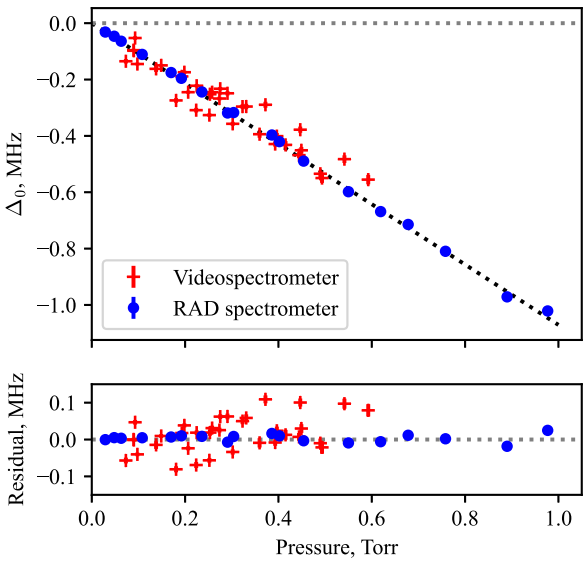

Fig. 3. Similar to Fig. 2 but for central frequency shift Δ_0.

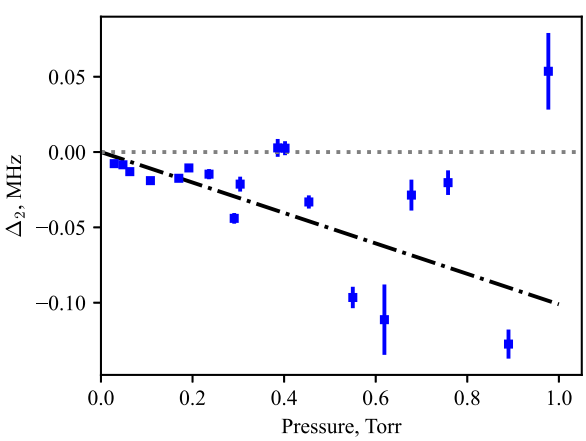

Fig. 4. Speed-dependent parameter Δ_2 of self-shifting of 752-GHz line versus pressure (RAD spectrometer data). Solid line is linear approximation of experimental points. Error bars show 1σ fit uncertainty.

unstable in terms of sign due to the correlation of the line shape and baseline parameters in the spectrometer model function. However, it was still possible to apply a linear regression to the data and also to retrieve δ_2 parameter from the multifit. When analyzing the video spectrometer recordings, we fixed Δ_2 to zero.

The line shape parameters retrieved from the video and RAD spectrometers data treatment were averaged with the weight defined by the fit uncertainty (except for δ_2 parameter which was only available from the RAD spectrometer measurements), and the total error was calculated accounting for the pressure and temperature readings uncertainty similar to our previous studies (e.g.,see eq. (18) from [27]). The resulting values of the broadening and shifting coefficients for both speed-independent and speed-dependent model profiles and for different perturbing molecules are given in Table 2.

3. Discussion

Self- and air-pressure broadening and shifting coefficients obtained using the Voigt line profile are shown in Table 3 together with the literature data from [5,6,9,11,17,29–31]. For the air collisional parameters, the agreement is quite good, while the self-parameters from other sources differ more from each other and from the results of this study. It is worth noting that our measured self-broadening for the 620-GHz line is about twice smaller than the HITRAN2020 value. The latter originates from the measurements by Podobedov et al. [8], where the full widths at half-maximum (FWHM) were presented in Table 1 in contrast to HWHM in HITRAN. Thus, the observed double difference is physically meaningless, it is just the result of incorrect inclusion of the original value into the database (the same is true for the lines at 30.6, 32.4, 42.6, 51.4 cm^{-1}). In the up-to-date version (HITRAN2024) [32] broadening coefficients of the corresponding lines are specified properly.

The measured parameters of the water lines were verified using a laboratory spectrum of a real atmosphere registered with the resonator spectrometer at atmospheric pressure and room temperature [9].

The absorption coefficient of the humid atmospheric air in the considered frequency range was a sum of the resonant lines of water vapor and oxygen (shown separately in the upper part of Fig. 5) and the water-related continuum absorption. Pure nitrogen absorption was subtracted from the total absorption together with the spectrometer baseline recorded in pure N_2. The pure oxygen continuum was negligibly small.

The resonant absorption was modeled using the Van Vleck–Weisskopf profile or its quadratic speed-dependent analogue [33] (if SD-parameters are available). These line shapes were derived in the impact approximation and thus are correct only in the vicinity of the line center. Consistent with the MT_CKD approach [34], the line wings were truncated at 25 cm^{-1} (750 GHz) frequency detuning from the line center and then the lines were "reduced" for the subsequent correct accounting of the continuum absorption modeled using the described approach. The wings beyond the cut-off 25 cm^{-1} and the plinth were attributed to the continuum. Lines in the range up to 1250 GHz were considered, as their wings can contribute to the absorption in the studied frequency range. Three datasets of the line shape parameters were tested: (i) HITRAN data solely [11]; (ii) HITRAN data supplemented with our experimental results; and (iii) PyRTlib line list [18].

The water-related continuum was considered as the sum of the self- and foreign-continuum. In the millimeter wave spectral range, the PWR model (the successor of MPM [35]) is traditionally used. This model utilizes the empirical quadratic frequency dependence and power law for the temperature dependence for both continuum components. The latest version, PWR2024 [36], gives the absorption magnitude very close to the more recent physically sound continuum model [37] developed based on the bimolecular interpretation of its physical origin, in particular, the spectral contributions of stable and metastable dimers supplemented with the correct approximations of the resonance line far wings. In the considered frequency range, the difference between these continuum models under the experimental conditions (room temperature, atmospheric pressure) is less than 1.5×10^{-7} cm^{-1}, which is comparable with the experimental noise and can be ignored for the validation purposes of the resonance part of absorption.

As is seen from Fig. 5, the absorption modeled with the HITRAN and PyRTlib parameters systematically differs from the experimental data. The form of these differences allows us to determine their primary sources. Thus, the shape of the residual near the 380-GHz line indicates an underestimation of intensity in both HITRAN2020 and PyRTlib which uses the value from HITRAN2004. The residual for the 448-GHz line has a shape of the first derivative of the line profile, which means that the incorrect pressure shift is tabulated for this line in the database. The correction of the 380-GHz line intensity by 2.5% and air-shifting parameter of the 448-GHz line by 1.7 times leads to a zero-biased W-shaped residual for both lines (see Fig. 5), indicating a further need to account for the SD effect. A similar behavior is observed for the model with the PyRTlib parameters. It is worth noting that the corrected air-shift parameter −0.152 MHz/Torr of the 448-GHz line is much closer to our experimental result −0.134 MHz/Torr than the initial value −0.0895 MHz/Torr. The use of the quadratic speed dependent Van Vleck–Weiskopf profile with the corresponding collisional parameters

Table 2
Experimental parameters of the studied lines. Values corresponding to dry air are calculated using 0.21, 0.78 and 0.01 for O_2, N_2 and Ar partial concentrations, respectively. Errors in parentheses are total uncertainties including fit errors together with the pressure and temperature readings uncertainty.

Central freq.	Parameter	Perturbing molecule				
MHz	MHz/Torr	H_2O	N_2	O_2	Ar	air
325152.896(9)	δ_V	1.704(29)	−0.036(1)	−0.013(1)	−0.027(1)	−0.031(1)
	γ_V	18.066(94)	4.112(24)	2.291(15)	1.750(11)	3.707(22)
	δ_0	1.767(13)	−0.056(3)	−0.011(2)	−0.026(5)	−0.046(3)
	δ_2	0.188(30)	−0.047(6)	—	0.004(9)	−0.037(5)
	γ_0	18.264(96)	4.242(25)	2.316(17)	1.775(13)	3.814(23)
	γ_2	1.772(127)	0.615(05)	0.207(5)	0.195(9)	0.525(5)
380197.356(5)	δ_V	0.313(47)	−0.104(1)	−0.081(1)	−0.104(3)	−0.099(1)
	γ_V	18.59(9)	4.206(25)	2.403(24)	1.812(19)	3.899(21)
	δ_0	0.416(56)	−0.120(2)	−0.085(2)	−0.103(5)	−0.112(2)
	δ_2	0.25(12)	−0.050(4)	−0.013(5)	−0.002(5)	−0.042(4)
	γ_0	18.72(15)	4.322(20)	2.421(26)	1.834(17)	3.804(25)
	γ_2	1.79(34)	0.607(8)	0.192(13)	0.172(5)	0.516(9)
439150.799(6)	δ_V	0.227(1)	0.111(7)	−0.038(5)	−0.166(1)	0.077(7)
	γ_V	11.997(64)	3.137(17)	1.405(18)	0.817(16)	2.751(17)
	δ_0	0.190(1)	0.107(3)	−0.046(2)	−0.168(1)	0.072(3)
	δ_2	−0.116(5)	0.005(2)	0.006(2)	—	0.005(2)
	γ_0	12.254(63)	3.178(18)	1.415(17)	0.856(17)	2.786(18)
	γ_2	1.251(14)	0.325(16)	0.116(6)	0.155(1)	0.280(14)
448001.082(6)	δ_V	−0.842(10)	−0.136(1)	−0.125(2)	−0.229(3)	−0.135(1)
	γ_V	17.665(91)	3.825(22)	2.085(13)	1.447(24)	3.437(20)
	δ_0	−0.841(6)	−0.150(1)	−0.130(1)	−0.221(4)	−0.146(1)
	δ_2	−0.018(7)	−0.031(1)	—	0.043(10)	−0.024(1)
	γ_0	17.96(1)	3.909(30)	2.112(12)	1.497(16)	3.509(26)
	γ_2	1.704(41)	0.550(5)	0.199(17)	0.211(3)	0.473(7)
474689.116(4)	δ_V	−1.239(66)	−0.036(07)	−0.100(02)	−0.229(03)	−0.051(6)
	γ_V	14.83(11)	3.485(24)	1.759(10)	1.115(21)	3.100(21)
	δ_0	−1.020(011)	−0.025(01)	−0.091(02)	−0.216(01)	−0.041(1)
	δ_2	—	—	—	—	—
	γ_0	15.079(104)	3.554(23)	1.776(11)	1.179(22)	3.158(20)
	γ_2	1.192(094)	0.426(07)	0.165(02)	0.217(03)	0.369(6)
556935.987(3)	δ_V	−2.205(15)	0.273(3)	0.126(4)	0.119(1)	0.241(3)
	γ_V	19.85(11)	4.689(41)	2.558(35)	1.683(15)	4.213(39)
	δ_0	−2.226(21)	0.257(3)	0.125(6)	0.118(3)	0.228(4)
	δ_2	−0.349(21)	−0.021(11)	0.018(4)	−0.017(3)	−0.013(9)
	γ_0	20.15(11)	4.89(4)	2.612(30)	1.760(18)	4.382(10)
	γ_2	1.47(11)	0.801(6)	0.252(14)	0.316(20)	0.681(8)
620700.952(2)	δ_V	−1.577(16)	−0.071(6)	−0.057(2)	−0.211(2)	−0.069(5)
	γ_V	15.52(12)	3.601(19)	1.726(10)	0.984(15)	3.182(17)
	δ_0	−1.607(17)	−0.074(9)	−0.061(3)	−0.198(2)	−0.072(8)
	δ_2	−0.141(13)	—	—	0.019(2)	—
	γ_0	15.93(9)	3.717(21)	1.763(10)	1.144(11)	3.282(19)
	γ_2	1.809(86)	0.458(4)	0.183(8)	0.279(12)	0.399(5)
752033.110(2)	δ_V	−1.056(10)	0.223(2)	0.127(1)	0.153(2)	0.202(2)
	γ_V	17.65(12)	4.548(27)	2.466(32)	1.760(36)	4.084(28)
	δ_0	−1.095(10)	0.219(2)	0.125(1)	0.140(2)	0.198(2)
	δ_2	−0.130(11)	—	—	−0.029(3)	—
	γ_0	17.99(14)	4.723(35)	2.586(44)	1.856(31)	4.247(37)
	γ_2	2.070(33)	0.835(9)	0.287(13)	0.264(9)	0.715(10)
859965.637(3)	δ_V	−1.256(7)	0.180(7)	0.168(33)	0.242(4)	0.178(12)
	γ_V	16.57(9)	4.34(11)	2.389(26)	1.555(19)	3.904(14)
	δ_0	−1.298(33)	0.157(12)	—	0.312(87)	0.161(17)
	δ_2	−0.249(5)	—	—	—	—
	γ_0	17.01(9)	4.555(89)	—	1.569(41)	3.571(70)
	γ_2	2.056(12)	0.681(19)	—	0.289(76)	0.534(16)

Table 3
Comparison of room-temperature Voigt profile pressure-broadening and pressure-shifting parameters with previous data.

Frequency	Parameter (MHz/Torr)				Reference
(MHz)	γ_V self	δ_V self	γ_V air	δ_V air	
325152.896(9)	18.066(94)	1.704(29)	3.707(23)	−0.031(1)	this study
325152.937(30)	20.0(40)	–	3.95(40)	−0.1089[a]	[11]
	19.556	1.462	–	–	[15][b]
325152.888(5)	18.51(3)	1.764(8)	3.78(3)	−0.017(6)	[29]
380197.356(5)	18.59(9)	0.313(47)	3.804(26)	−0.099(1)	this study
380197.418(30)	25.1(>50)	–	3.84(38)	−0.1278[a]	[11]
	20.597	0.247	–	–	[15][b]
380197.353(6)	19.20(7)	0.32(4)	3.774(8)	−0.105(5)	[9]
380197.365(5)	19.5(1)	0.315(2)	3.83(5)	−0.094(6)	[29]
439150.799(6)	11.997(64)	0.227(1)	2.751(18)	0.077(7)	this study
439150.864(30)	13.0(13)	–	2.82(56)	0.0911[a]	[11]
	12.514	−0.199	–	–	[15][b]
439150.791(10)	13.06(25)	0.346(10)	–	–	[5]
439150.807(9)	12.08(5)	0.220(46)	2.699(33)	0.049(27)	[9]
448001.082(6)	17.665(91)	−0.842(10)	3.436(22)	−0.134(1)	this study
448001.128(30)	16.3(33)	–	3.5(7)	−0.0895[a]	[11]
	18.414	−1.135	–	–	[15][b]
448001.085(10)	17.34(3)	−0.820(35)	3.449(6)	−0.160(3)	[9]
474689.116(4)	14.83(11)	−1.239(66)	3.100(23)	−0.051(6)	this study
474689.166(30)	15.6(31)	–	3.16(63)	−0.0024[a]	[11]
	15.611	−1.377	–	–	[15][b]
474689.092(8)	14.98(6)	−0.96(10)	3.12(3)	−0.06(3)	[9]
556935.987(3)	19.85(11)	−2.205(15)	4.212(41)	0.240(3)	this study
556936.069(30)	18.9(38)	–	4.35(22)	0.2584[a]	[11]
	19.192	−2.233	–	–	[15][b]
556935.985(2)	–	–	4.92(6)	0.307(4)	[6]
	18.8(2)	−2.235(100)	4.14(2)	0.25(2)	[30]
620700.952(2)	15.52(12)	−1.577(16)	3.182(18)	−0.069(5)	this study
620701.035(30)	31.87(32)	–	3.46(70)	0.0201[a]	[11]
	15.78(6)	1.815(79)	–	–	[8]
	16.519	−1.271	–	–	[15][b]
752033.110(2)	17.65(12)	−1.056(10)	4.084(31)	0.202(2)	this study
752033.234(30)	18.0(36)	–	4.23(21)	0.1905[a]	[11]
	18.267	−1.38	–	–	[15][b]
752033.113(14)	18.11(15)	−1.17(3)	4.152(46)	0.216(27)	[31]
859965.637(3)	16.57(9)	–	3.903(16)	0.178(13)	this study
859965.7(30)	18.776(—)	–	4.16(83)	0.1905[a]	[11]

[a] Uncertainty of the HITRAN air-shifting is within 0.03–0.3 MHz (code 6).
[b] Semi-classical calculations results.

measured in the current study and the HITRAN intensities (with the aforementioned corrections) allowed reducing the residual down to the noise level, proving better modeling accuracy of the resonance absorption.

The root mean square (RMS) of the measured minus the calculated difference for the data from HITRAN, PyRTlib and current study are 11.6×10^{-7}, 7.6×10^{-7} and 4.7×10^{-7} cm^{-1}, respectively. The corrected HITRAN data gives the RMS value of 6.3×10^{-7} cm^{-1}.

4. Conclusions

In the range below 1 THz, a large amount of experimental recordings of the key water vapor lines was recorded at room temperature and analyzed to retrieve a set of collisional parameters for modeling the Earth's atmosphere absorption coefficient, demonstrating high accuracy. The use of three fundamentally different spectrometers allowed both accurate measurement of the line-shape parameters and their independent validation, thus increasing reliability of the obtained data. Some inaccuracies in the HITRAN data were demonstrated, while the proposed dataset on line shape parameters, including SD coefficients, is able to significantly improve modeling accuracy.

CRediT authorship contribution statement

M.A. Koshelev: Writing – review & editing, Supervision, Methodology, Conceptualization. **I.N. Vilkov:** Methodology, Investigation. **D.S. Makarov:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation. **G.Yu. Golubiatnikov:** Investigation, Data curation. **A.O. Koroleva:** Writing – review & editing, Visualization, Validation, Software, Data curation. **T.A. Galanina:** Writing – review & editing, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The study was supported by Russian Science Foundation, Russia (project no. 22–72–10118 https://rscf.ru/en/project/22-72-10118/). Experimental data were obtained using the research facilities of "CKP-7" (USU No 3589084).

Data availability

Data will be made available on request.

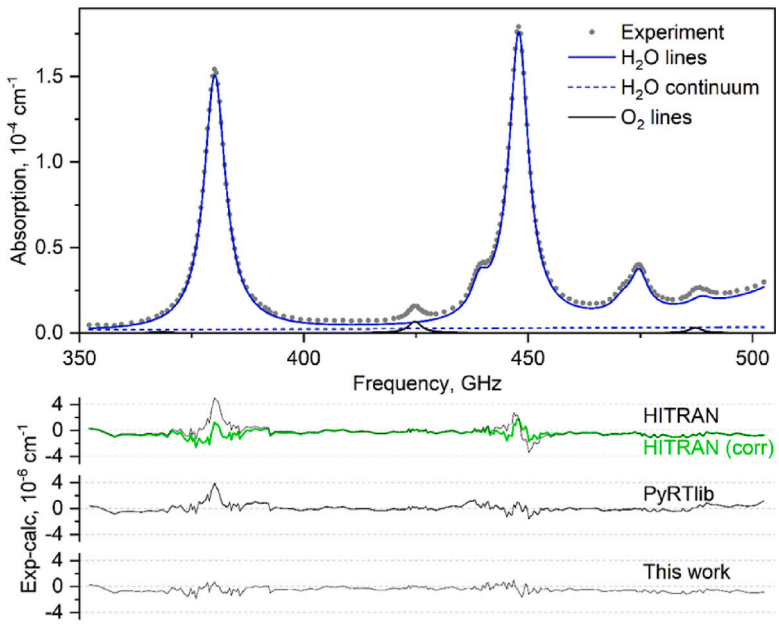

Fig. 5. Comparison of the measured absorption coefficient (gray points) of the atmospheric air at pressure 758.5 Torr, temperature 296 K and water content of 1.667 Torr with the calculated one (see details in the inset). Exp-minus-calc residuals are shown in the lower part of the figure for HITRAN2020 [11] (upper trace, black curve) and its corrected version (upper trace, green curve), PyRTlib [18] (central trace) and current study (lower trace) datasets.

References

[1] Gallucci D, Cimini D, Turner E, Fox S, Rosenkranz PW, Tretyakov MY, Mattioli V, Larosa S, Romano F. Uncertainty in simulated brightness temperature due to sensitivity to atmospheric gas spectroscopic parameters from the centimeter- to submillimeter-wave range. Atmos Chem Phys 2024;24(12):7283–308.

[2] Tretyakov MY. Spectroscopy underlying microwave remote sensing of atmospheric water vapor. J Molec Spectrosc 2016;328:7–26.

[3] Koshelev MA, Golubyatnikov GY, Vilkov IN, Tretyakov MY. Line shape parameters of the 22-GHz water line for accurate modeling in atmospheric applications. J Quant Spectrosc Radiat Transfer 2018;205:51–8.

[4] Koshelev MA, Vilkov IN, Makarov DS, Tretyakov MY, Vispoel B, Gamache RR, Cimini D, Romano F, Rosenkranz PW. Water vapor line profile at 183-GHz: Temperature dependence of broadening, shifting, and speed-dependent shape parameters. J Quant Spectrosc Radiat Transfer 2021;262:107472.

[5] Markov VN. Temperature dependence of self-induced pressure broadening and shift of the 6_{43}–5_{50} line of the water molecule. J Molec Spectrosc 1994;164(1):233–8.

[6] Markov VN, Krupnov AF. Measurements of the pressure shift of the 1_{10}–1_{01} water line at 556 GHz produced by mixtures of gases. J Molec Spectrosc 1995;172(1):211–4.

[7] Ryadov VY, Furashov NI. The width of the water vapor absorption line $\lambda = 0.92$ mm. Sov Radiophys 1969;9(6):621–3.

[8] Podobedov VB, Plusquellic DF, Fraser GT. THz laser study of self-pressure and temperature broadening and shifts of water vapor lines for pressures up to 1.4 kPa. J Quant Spectrosc Radiat Transfer 2004;87(3):377–85.

[9] Tretyakov M, Koshelev M, Vilkov I, Parshin V, Serov E. Resonator spectroscopy of the atmosphere in the 350–500 GHz range. J Quant Spectrosc Radiat Transfer 2013;114:109–21.

[10] Mlawer EJ, Turner DD, Paine SN, Palchetti L, Bianchini G, Payne VH, Cady-Pereira KE, Pernak RL, Alvarado MJ, Gombos D, Delamere JS, Mlynczak MG, Mast JC. Analysis of water vapor absorption in the far-infrared and submillimeter regions using surface radiometric measurements from extremely dry locations. JGR Atmos 2019;124(14):8134–60.

[11] Gordon IE, Rothman LS, Hargreaves RJ, Hashemi R, Karlovets EV, Skinner FM, Conway EK, Hill C, Kochanov RV, Tan Y, o PW, Finenko AA, Nelson K, Bernath PF, Birk M, Boudon V, Campargue A, Chance KV, Coustenis A, Drouin BJ, Flaud J-M, Gamache RR, Hodges JT, Jacquemart D, Mlawer EJ, Nikitin AV, Perevalov VI, Rotger M, Tennyson J, Toon GC, Tran H, Tyuterev VG, Adkins EM, Baker A, Barbe A, Canè E, Császár AG, Dudaryonok A, Egorov O, Fleisher AJ, Fleurbaey H, Foltynowicz A, Furtenbacher T, Harrison JJ, Hartmann J-M, Horneman V-M, Huang X, Karman T, Karns J, Kassi S, Kleiner I, Kofman V, Kwabia-Tchana F, Lavrentieva N, Lee T, Long D, Lukashevskaya A, Lyulin O, Makhnev V, Matt W, Massie S, Melosso M, Mikhailenko S, Mondelain D, Müller H, Naumenko O, Perrin A, Polyansky O, Raddaoui E, Raston P, Reed Z, Rey M, Richard C, Tóbiás R, Sadiek I, Schwenke D, Starikova E, Sung K, Tamassia F, Tashkun S, Vander Auwera J, Vasilenko I, Vigasin A, Villanueva G, Vispoel B, Wagner G, Yachmenev A, Yurchenko S. The HITRAN2020 molecular spectroscopic database. J Quant Spectrosc Radiat Transfer 2022;277:107949.

[12] Antony BK, Neshyba S, Gamache RR. Self-broadening of water vapor transitions via the complex Robert–Bonamy theory. J Quant Spectrosc Radiat Transfer 2007;105(1):148–63.

[13] Jacquemart D, Gamache R, Rothman LS. Semi-empirical calculation of air-broadened half-widths and air pressure-induced frequency shifts of water-vapor absorption lines. J Quant Spectrosc Radiat Transfer 2005;96(2):205–39.

[14] Vispoel B, Cavalcanti JH, Paige ET, Gamache RR. Vibrational dependence, temperature dependence, and prediction of line shape parameters for the H_2O–N_2 collision system. J Quant Spectrosc Radiat Transfer 2020;253:107030.

[15] Cazzoli G, Puzzarini C, Buffa G, Tarrini O. Pressure-broadening of water lines in the THz frequency region: Improvements and confirmations for spectroscopic databases.: Part I. J Quant Spectrosc Radiat Transfer 2008;109(17):2820–31.

[16] Cazzoli G, Puzzarini C, Buffa G, Tarrini O. Pressure-broadening of water lines in the THz frequency region: Improvements and confirmations for spectroscopic databases. Part II. J Quant Spectrosc Radiat Transfer 2009;110(9):609–18.

[17] Cazzoli G, Puzzarini C, Buffa G, Tarrini O. Pressure-broadening in the THz frequency region: The 1.113 THz line of water. J Quant Spectrosc Radiat Transfer 2008;109(9):1563–74.

[18] Larosa S, Cimini D, Gallucci D, Nilo S, Romano F. PyRTlib: an educational Python-based library for non-scattering atmospheric microwave radiative transfer computations. Geosci Model Dev 2024;17:2053–76.

[19] Golubiatnikov GY, Belov SP, Leonov II, Andriyanov AF, Zinchenko II, Lapinov AV, Markov VN, Shkaev AP, Guarnieri A. Precision sub-Doppler millimeter and submillimeter lamb-dip spectrometer. Radiophys. Quantum El. 2014;56:599–609.

[20] Golubiatnikov GY, Belov SP, Lapinov AV. The features of the frequency-modulation method when studying the shapes of the spectral lines of nonlinear absorption. Radiophys. Quantum El. 2017;59:715–26.

[21] Krupnov AF. In: Chantry G, editor. Modern aspects of microwave spectroscopy. Academic Press; 1979, p. 217–56, L.

[22] Tretyakov MY, Koshelev MA, Makarov DS, Tonkov MV. Precise measurements of collision parameters of spectral lines with a spectrometer with radioacoustic detection of absorption in the millimeter and submillimeter ranges. Instrum Exp Tech 2008;51(1):78–88.

[23] Koshelev MA, Tsvetkov AI, Morozkin MV, Glyavin MY, Tretyakov MY. Molecular gas spectroscopy using radioacoustic detection and high-power coherent subterahertz radiation sources. J Molec Spectrosc 2017;331:9–16.

[24] Golubiatnikov GY, Koshelev MA, Tsvetkov AI, Fokin AP, Glyavin MY, Tretyakov MY. Sub-terahertz high-sensitivity high-resolution molecular spectroscopy with a gyrotron. IEEE Trans Terahertz Sci Technol 2020;10(5):502–12.

[25] Koshelev MA, Leonov II, Serov EA, Chernova AI, Balashov AA, Bubnov GM, Andriyanov AF, Shkaev AP, Parshin VV, Krupnov AF, Tretyakov MY. New frontiers in modern resonator spectroscopy. IEEE Trans Terahertz Sci Technol 2018;8:773–83.

[26] Ngo NH, Lisak D, Tran H, Hartmann JM. An isolated line-shape model to go beyond the Voigt profile in spectroscopic databases and radiative transfer codes. J Quant Spectrosc Radiat Transfer 2013;129:89–100.

[27] Tretyakov MY, Serov EA, Makarov DS, Vilkov IN, Golubiatnikov GY, Galanina TA, Koshelev MA, Balashov AA, Simonova AA, Thibault F. Pure rotational R(0) and R(1) lines of CO in Ar baths: experimental broadening, shifting and mixing parameters in a wide pressure range versus *ab initio* calculations. Phys. Chem Chem Phys 2023;25:1310–30.

[28] Makarov DS, Vilkov IN, Galanina TA, Serov EA, Koshelev MA, Tretyakov MY. Collisional parameters of the pure rotational R(0) line of CO in N_2 : Experiment versus theory. J Quant Spectrosc Radiat Transfer 2025;337:109379.

[29] Koshelev MA, Tretyakov MY, Golubiatnikov GY, Parshin VV, Markov VN, Koval IA. Broadening and shifting of the 321-, 325- and 380-GHz lines of water vapor by pressure of atmospheric gases. J Molec Spectrosc 2007;241:101–8.

[30] Golubiatnikov GY, Koshelev MA, Krupnov AF. Pressure shift and broadening of $1_{10}-1_{01}$ water vapor lines by atmosphere gases. J Quant Spectrosc Radiat Transfer 2008;109:1828–33.

[31] Koshelev MA. Collisional broadening and shifting of the $2_{11} - 2_{02}$ transition of $H_2^{16}O$, $H_2^{17}O$, $H_2^{18}O$ by atmosphere gases. J Quant Spectrosc Radiat Transfer 2011;112:550–2.

[32] Gordon I, Rothman L, Hargreaves R, Gomez F, Bertin T, Hill C, Kochanov R, Tan Y, Wcisło P, Makhnev VY, Bernath P, Birk M, Boudon V, Campargue A, Coustenis A, Drouin B, Gamache R, Hodges J, Jacquemart D, Mlawer E, Nikitin A, Perevalov V, Rotger M, Robert S, Tennyson J, Toon G, Tran H, Tyuterev V, Adkins E, Barbe A, Bailey D, Bielska K, Bizzocchi L, Blake T, Bowesman C, Cacciani P, Čermák P, Császár A, Denis L, Egbert S, Egorov O, Ermilov AY, Fleisher A, Fleurbaey H, Foltynowicz A, Furtenbacher T, Germann M, Guest E, Harrison J, Hartmann J-M, Hjältén A, Hu S-M, Huang X, Johnson T, Jóźwiak H, Kassi S, Khan M, Kwabia-Tchana F, Lee T, Lisak D, Liu A-W, Lyulin O, Malarich N, Manceron L, Marinina A, Massie S, Mascio J, Medvedev E, Meshkov V, Mellau GC, Melosso M, Mikhailenko S, Mondelain D, Müller H, O'Donnell M, Owens A, Perrin A, Polyansky O, Raston P, Reed Z, Rey M, Richard C, Rieker G, Röske C, Sharpe S, Starikova E, Stolarczyk N, Stolyarov A, Sung K, Tamassia F, Terragni J, Ushakov V, Vasilchenko S, Vispoel B, Vodopyanov K, Wagner G, Wójtewicz S, Yurchenko S, Zobov N. The HITRAN2024 molecular spectroscopic database. J Quant Spectrosc Radiat Transfer 2026;109807.

[33] Koshelev MA, Delahaye T, Serov EA, Vilkov IN, Boulet C, Tretyakov MY. Accurate modeling of the diagnostic 118-GHz oxygen line for remote sensing of the atmosphere. J Quant Spectrosc Radiat Transfer 2017;196:78–86.

[34] Mlawer K, Cady-Pereira E, Mascio J, Gordon IE. The inclusion of the MT_CKD water vapor continuum model in the HITRAN molecular spectroscopic database. J Quant Spectrosc Radiat Transfer 2023;306:108645.

[35] Liebe H, Gimmestad G. Calculation of clear air EHF refractivity. Radio Sci 1978;13(2):245–51.

[36] Rosenkranz PW. Line-by-line microwave radiative transfer (non-scattering) [software] (version 2024/07/03). URL http://cetemps.aquila.infn.it/mwrnet/lblmrt_ns.html.

[37] Tretyakov MY, Galanina TA, Koroleva AO, Makarov DS, Chistikov DN, Finenko AA, Vigasin AA. Atmospheric water vapor continuum model for the sub-THz range. J Quant Spectrosc Radiat Transfer 2025;333:109319.